

Lecture 22 – Electro-magnetic Induction

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We will

- examine experimental evidence that a changing magnetic field induces an emf
- learn how Faraday's law relates the induced emf to the change in flux
- determine the direction of an induced emf
- calculate the emf induced by a moving conductor
- learn how a changing magnetic flux generates an electric field

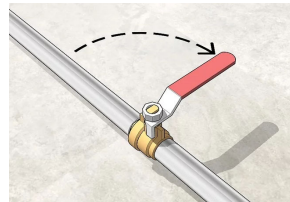
A few slides in this chapter come from Chapter-29 of Young's textbook.

1

Electromagnetic Induction



Water Hammer



When you abruptly turn on (or turn off) your water pipes made of plastic material, you can always vividly see the hydraulic shock, and the shock wave goes through your pipes, forcing your pipes to move, shake, and you can also hear the noise of bang together. Even for metal pipes you can at least hear the noise. This is potentially dangerous and could hurt your plumbing system. That explains why some families have more frequent water leakages from their pipes than others. What I can advise you is to be gentle when you turn on (or turn off) your water taps so as to reduce such water-hammering damages to your plumbing system at home. In this chapter I will be taking about potential difference and accordingly there will be current and electric field. But these potential difference, current and electric field are not due to the energy coming from battery. In fact there will be no battery in our derivations. These potential difference, current and electric field are induced, thus its electric field is **not** conservative!!!

- A/Prof Tay

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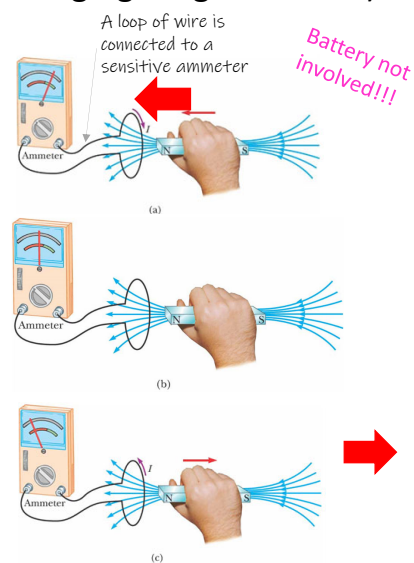


Water Hammering Video

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Experimental Observations in Physics Laboratory (Induced EMF Produced by a Changing Magnetic Field):

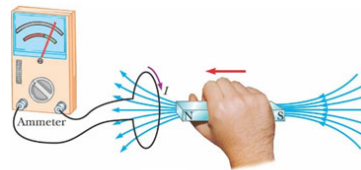
- When a magnet is moved toward the loop, the ammeter deflects in one direction (for the convenience of discussion we place the pointer to the right)
- When the magnet is held stationary, there is no deflection of the ammeter. Therefore, there is no induced current even though the magnet is in the loop
- When the magnet is moved away from the loop, the ammeter deflects in the opposite direction



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Induction

A current can be produced **without** a battery present in the circuit. The ammeter deflects due to the **induction of current**.



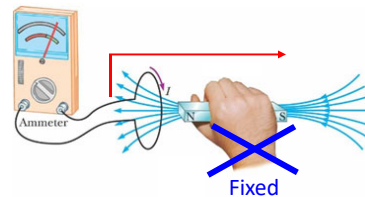
The current is **induced** because there isn't any battery!!

- An **induced current** is produced by a **changing (increasing or decreasing)** magnetic field
- As there is an **induced current**, there must be an **induced e.m.f.** associated with the induced current
(because without a difference in electric potential, electric current will not flow.)
(... without a difference in height, water will not flow.)

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EMF Produced by a Changing Magnetic Field, Summary

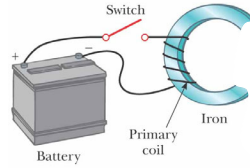
- The ammeter deflects when the magnet is moving toward or away from the loop
- The ammeter also deflects when the loop is moved toward or away from the magnet
- Therefore, the loop detects that the magnet is moving relative to it. We relate this detection to a change in the magnetic field. This is the induced current that is produced by the induced e.m.f.



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Faraday's Experiment

- A primary coil is connected to a switch and a battery, and the wire is wrapped around one side of an iron ring (soft magnetic material)
- A secondary coil is also wrapped around the iron ring on the opposite side of primary coil, and it is connected to the ammeter
- There is no battery present in the secondary coil
- The secondary coil is not directly connected to the primary coil
- At the instant the switch is closed, the ammeter needle deflects in one direction and then returns to zero
- The ammeter continues to read zero even when there is a steady current in the primary circuit
- When the switch is opened, the ammeter needle deflects in the opposite direction and then returns to zero
- The ammeter continues to read zero when there is no current in the primary circuit



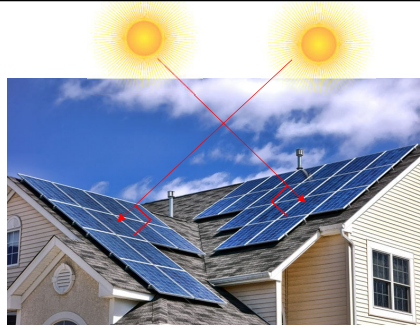
No battery here

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Magnetic Flux

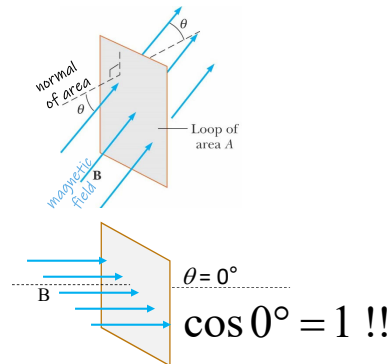
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A solar **panel** will harness the most power when the Sun's rays hit its surface perpendicularly.



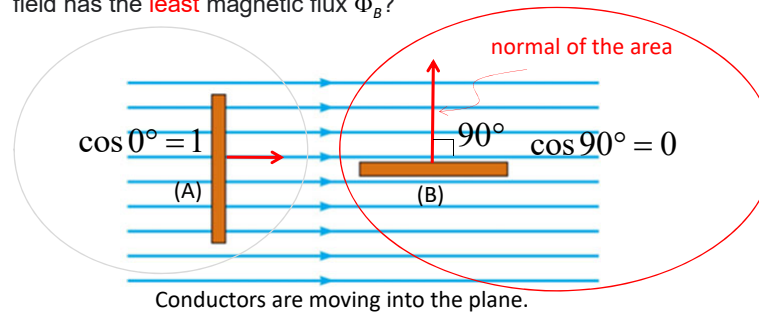
Magnetic Flux

- Assume a loop enclosing an area A lies in a uniform magnetic field $\mathbf{B}$, the magnetic flux Φ_B through the loop is defined as $\Phi_B = BA \cos \theta$ where θ is the angle between the normal of the area and the magnetic field
- The magnetic flux through the loop is maximum when $\theta = 0^\circ$, i.e., when the magnetic field penetrates the loop at 90° .



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Example. Which of the following moving conductors in a magnetic field has the **least** magnetic flux Φ_B ?



Conductors are moving into the plane.

Answer:

$$\Phi_B = BA \cos \theta$$

where θ is the angle between the normal of the area and the magnetic field.

- (A) When the field is perpendicular to the area, $\theta = 0^\circ$ and $\Phi = \Phi_{\max} = BA$

- (B) When the field is parallel to the area, $\theta = 90^\circ$ and $\Phi = 0$

Unit for Magnetic Flux Φ_B

$$\Phi_B = BA \cos \theta$$

$$\text{T} \times \text{m}^2 = \text{Weber (Wb)}$$

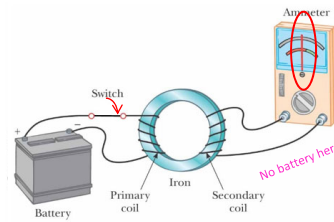
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Faraday's Experiment – Conclusions

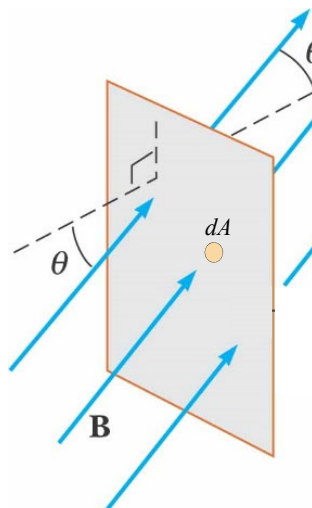
- An electric current can be induced in a circuit by a changing magnetic field. This would be the current in the secondary circuit.
- The induced current exists only for a short time while the magnetic field is changing. This is generally expressed as: ***an induced e.m.f. is produced in the secondary circuit by the changing magnetic field.***
- The actual existence of the magnetic flux is not sufficient to produce the induced e.m.f.; the flux must be **changing**.
- **Faraday's law of induction** states that "The e.m.f. $\mathcal{E}$ induced in a circuit is directly proportional to the time rate of change of the magnetic flux through the circuit." Mathematically:

$$\mathcal{E} = - \frac{d\Phi_B}{dt}$$

The minus sign expresses **Lenz's law**. It means that the induced e.m.f. is in such a direction that, if the circuit is closed (or is opened), the induced current **opposes** the change of magnetic flux. This means if the magnetic flux is increasing, then the induced current will generate a magnetic flux in the opposite direction, and vice versa.



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Faraday's Experiment – Conclusions, cont

From the magnetic flux $\Phi_B = BA \cos \theta$, what if the area dA in the figure is very small and is only part of a bigger area enclosed by a circuit?

Answer:

$$d\Phi_B = B dA \cos \theta$$

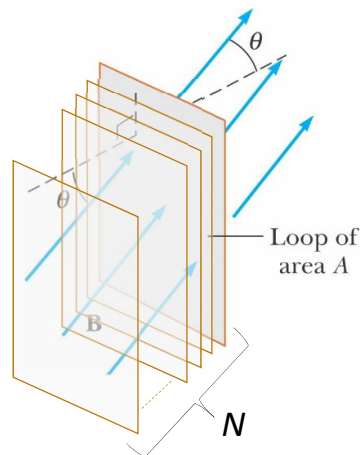
Remember Φ_B is the magnetic flux which penetrates the area enclosed by the whole circuit. So,

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A}$$

dot product

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Faraday's Experiment – Conclusions, cont



The magnetic flux on one loop is

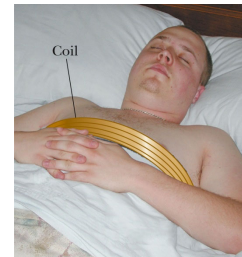
$$\Phi_B = BA \cos \theta$$

If the circuit consists of \$N\$ loops all of the same area, and \$\Phi_B\$ is the magnetic flux through one loop, the total e.m.f. \$\epsilon_{\text{total}}\$ induced in \$N\$ loops will be

$$\begin{aligned}\epsilon_{\text{total}} &= - \left(\frac{d\Phi_B}{dt} \right) \times N \\ &= -N \left(\frac{d}{dt} (BA \cos \theta) \right)\end{aligned}$$

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Example. To monitor the breathing of a hospital patient, a thin belt is girded around the patient's chest as shown. The belt is a 200-turn coil. When the patient inhales, the area encircled by the coil increases by \$39.0 \text{ cm}^2\$. The magnitude of the earth's magnetic field is \$50.0 \mu\text{T}\$ and makes an angle of \$28.0^\circ\$ with the plane of the coil. Assuming a patient takes \$1.80 \text{ s}\$ to inhale, find the magnitude of the average induced emf in the coil during that time interval.



Answer:

Magnitude of change of magnetic flux through one loop:

$$|\Delta\Phi| = B|\Delta A| \cos \theta$$

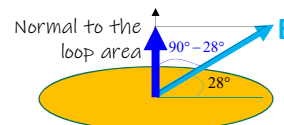
The angle \$\theta\$ is the angle between the magnetic field and the normal of the loop area.

$$\Rightarrow |\Delta\Phi| = (50.0 \times 10^{-6} \text{ T})(39.0 \times 10^{-4} \text{ m}^2) \cos(90.0^\circ - 28.0^\circ)$$

$$\Rightarrow |\Delta\Phi| = 9.15 \times 10^{-8} \text{ T} \cdot \text{m}^2$$

Average magnitude of induced emf:

$$|\epsilon| = N \frac{|\Delta\Phi|}{\Delta t} = (200) \frac{9.15 \times 10^{-8} \text{ T} \cdot \text{m}^2}{1.80 \text{ s}} = 1.02 \times 10^{-5} \text{ V}$$



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Example. A coil consists of 200 turns of wire. Each turn is a square of side 18 cm, and a uniform magnetic field directed perpendicular to the plane of the coil is turned on. If the field changes linearly from 0 to 0.50 T in 0.80 s, what is the magnitude of the induced e.m.f. in the coil while the field is changing?

Answer:

The area of one turn of the coil is $(0.18 \text{ m})^2 = 0.0324 \text{ m}^2$. The magnetic flux through the coil at $t = 0$ is zero because $B = 0$ at that time. At $t = 0.80 \text{ s}$, the magnetic flux through one turn is $\Phi_B = B \times A \times \cos \theta = (0.50 \text{ T})(0.0324 \text{ m}^2)(\cos 0^\circ) = 0.0162 \text{ T} \cdot \text{m}^2$

$$|\mathcal{E}| = N \frac{\Delta \Phi_B}{\Delta t} = 200 \times \frac{(0.0162 \text{ T} \cdot \text{m}^2 - 0)}{0.80 \text{ s}} = 4.1 \text{ T} \cdot \text{m}^2 / \text{s} = 4.1 \text{ V}$$

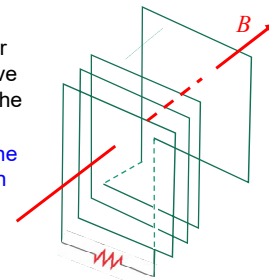
You should be able to show that $1 \frac{\text{T} \cdot \text{m}^2}{\text{s}} = 1 \text{ V}$

What If? What if you were asked to find the magnitude of the induced current in the coil while the field is changing? Can you answer this question?

Answer:

If the ends of the coil are not connected to a circuit, the answer to this question is easy – the current is zero! (Charges will move within the wire of the coil, but they cannot move into or out of the ends of the coil.) **In order for a induced current to exist, the ends of the coil must be connected to an external circuit. Let the total resistance of the coil and the circuit be $R \Omega$. The current in the coil where the ends are connected will be**

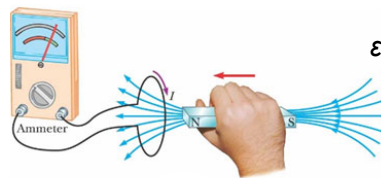
$$R = \frac{\Delta V}{I} \rightarrow I = \frac{\Delta V}{R} \quad I = \frac{\mathcal{E}}{R} = \frac{4.1 \text{ V}}{R \Omega} = \frac{4.1}{R} \text{ A}$$



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(inducing a potential difference!!)

Ways of Inducing an e.m.f.



$$\begin{aligned} \mathcal{E}_1 &= - \left(\frac{d}{dt} (BA \cos \theta) \right) \times 1 \text{ turn} \\ &= - \frac{d}{dt} (BA \cos \theta) \times n \text{ turns} \end{aligned}$$

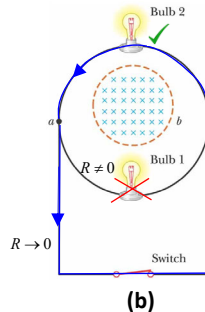
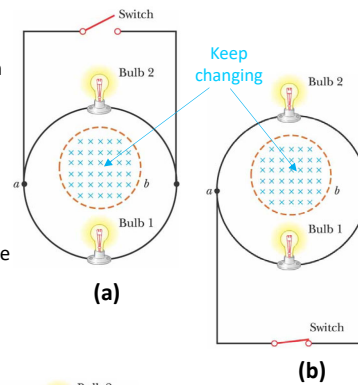
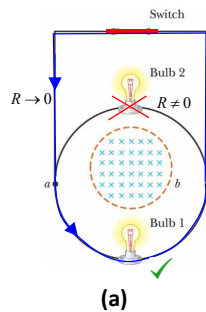
- The magnitude of magnetic field **B** can change with time
- The area **A** enclosed by the loop can change with time
- The angle θ between **B** and the normal to the loop can change with time
- Any combination of the above can occur

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Example. Two bulbs are connected to opposite sides of a circular loop of wire as shown in figure. A changing magnetic-field confined to the smaller circular area shown in the figure (a) induces an e.m.f. in the loop that causes the two bulbs to light. When the switch is connected, the inverted U-shape nearly-resistance-free wires connected to the switch short out Bulb 2 and only Bulb 1 remains lighted.

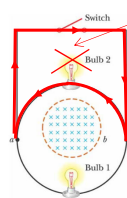
What happens if the wires containing the closed switch remain connected at points *a* and *b*, but the switch and the wires are lifted up and moved to the other side of the field, as in figure (b)? The wires are still connected to Bulb 2 as it was before, so does it continue to stay dark?

Answer:

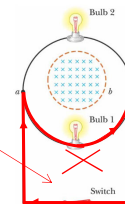


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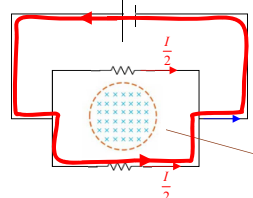
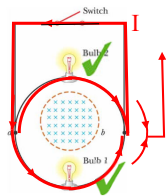
What if you **hard-connect** the red wire which has resistance?



The magnetic field in this enclosed area is 0 so there is no way to induce the current!!!



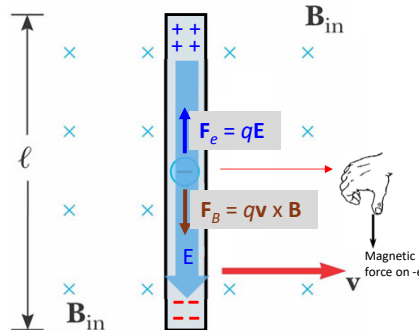
This is possible provided there is external e.m.f. (battery).



But in this example the source of e.m.f. is solely due to the changing magnetic flux. There is no battery.

Motional e.m.f.

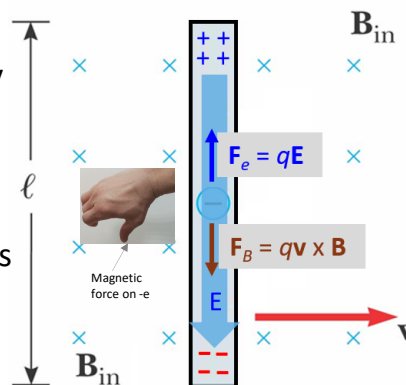
- A **motional e.m.f.** is the one induced in a conductor **moving** through a constant magnetic field
- The electrons in the conductor experience a magnetic force, $\mathbf{F}_B = q\mathbf{v} \times \mathbf{B}$ that is directed along ℓ . Under the influence of the magnetic force, the charge carriers (filled with electrons) move to the lower end of the conductor and accumulate there
- As a result of the charge separation (polarization), an electric field $\mathbf{E}$ is produced inside the conductor. (This induced electric field is not conservative). The charges accumulate at both ends of the conductor until they are in equilibrium with regard to the **electric force** $\mathbf{F}_e$ and **magnetic force** $\mathbf{F}_B$



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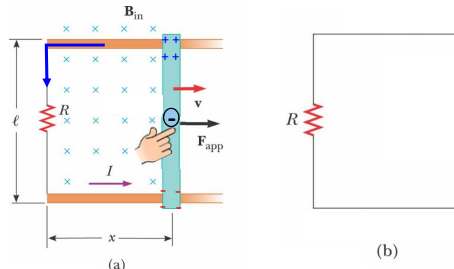
Motional e.m.f. , final

- For equilibrium, **magnetic force** $\mathbf{F}_B$ (downward) is balanced by **electric force** $\mathbf{F}_e$ (upward), i.e., $q\mathbf{v}B = qE$, or $E = vB$
- A potential difference will be maintained between the ends of the conductor **as long as** the conductor continues to move through the uniform magnetic field
- If the direction of the motion is reversed, the polarity of the potential difference will also be reversed



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What if the conductor is part of a circuit? Sliding Conducting Bar



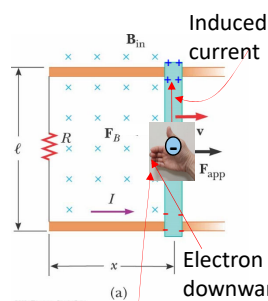
(a)
A bar moving through
a uniform field (by
your finger!!)

(b)
The equivalent
circuit diagram

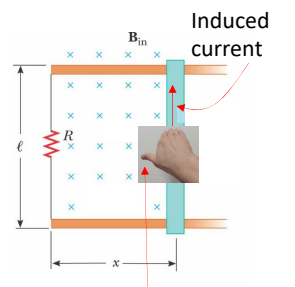
- Assume the bar has zero resistance
- The work done by the applied force $\mathbf{F}_{\text{app}}$ (from your finger) appears as internal energy in the resistor R

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Conducting bar moves to the right.
Area increasing.
Magnetic flux increasing.
Apply Lenz's law.



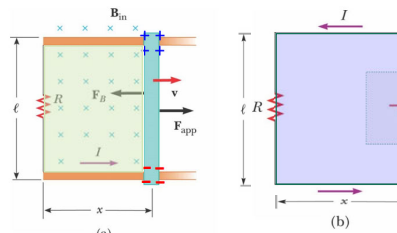
Electron is going
downward $\rightarrow$ induced
current is going
upward
Induced magnetic field is
going out (.) of the page
to oppose the **increasing**
inward (x) magnetic flux
(Lenz's Law)



Direction of
magnetic force on
the **conducting bar**
due to the **inward**
magnetic field (x)

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Sliding Conducting Bar, cont.



(a) A bar moving through a uniform field (by your finger!!)

(b) The equivalent circuit diagram

The induced e.m.f. is

$$\varepsilon = -\frac{d\Phi_B}{dt} = -\frac{d(B \times A \times \cos 0^\circ)}{dt}$$

$$= -\frac{d(B \times (\ell \times dx) \times 1)}{dt}$$

$$= -B\ell \left(\frac{dx}{dt} \right) = -B\ell v$$

Since the resistance in the circuit is R ,

the current on the circuit is $I = \frac{|\varepsilon|}{R} = \frac{B\ell v}{R}$

But this is **induced** current as there is no battery.

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Sliding Conducting Bar, Energy Considerations

- The magnetic force $\mathbf{F}_B$ exerted on the conducting bar is
 $\mathbf{F}_B = I \ell \mathbf{B}$
- The applied force $\mathbf{F}_{app}$ has to overcome the magnetic force $\mathbf{F}_B$, i.e., $\mathbf{F}_{app} = -\mathbf{F}_B$
- When the applied force does work on the conducting bar, the charges move through a magnetic field. The change in energy of the system is exhibited by the resistor (such as in term of heat) during some time interval must be equal to the transfer of energy into the system by work (from your finger).
- The **power** $\mathcal{P}$ input is equal to the rate at which energy is delivered to the resistor, i.e.,

$$\mathcal{P} = \frac{\text{energy}}{\text{time}} = \frac{\text{force} \times \text{distance}}{\text{time}} = \text{force} \times \text{velocity} = F_{app} v = (I \ell B) v$$

$$= \left(I \left(\frac{IR}{v} \right) \right) v = I^2 R = \left(\frac{\varepsilon}{R} \right)^2 R = \frac{\varepsilon^2}{R}$$

$$P = I^2 R$$

$$P = I \times \Delta V$$

$$P = \frac{\Delta V^2}{R}$$

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Angular Velocity $\left(\omega = \frac{d\theta}{dt}\right)$ versus Linear Velocity $\left(v = \frac{ds}{dt}\right)$

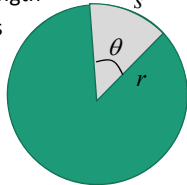
By definition: $\theta = \frac{s}{r}$

$\rightarrow d\theta = d\left(\frac{s}{r}\right) = \frac{ds}{r} = \left(\frac{1}{r}\right)ds$

$\frac{d\theta}{dt} = \left(\frac{1}{r}\right)\frac{ds}{dt}$

$\rightarrow \omega = \left(\frac{1}{r}\right)v$

$\rightarrow \mathbf{v = r\omega}$



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A conducting bar swinging in a circle

Example. A conducting bar of length ℓ rotates with a constant angular speed ω about a pivot at one end. A uniform magnetic field $\mathbf{B}$ is directed perpendicular to the plane of rotation, as shown in the figure. Find the motional e.m.f. induced between the ends of the bar.

Answer:

Consider a segment of the bar of length dr and width ds having a tangential velocity $\mathbf{v}$. The small e.m.f. $d\mathcal{E}$ due to the movement of the small area dA is:

$$d\mathcal{E} = \frac{B \times dA \times \cos 0^\circ}{dt} = \frac{B \times (ds \times dr) \times 1}{dt} = B \times \left(\frac{ds}{dt}\right) \times dr = Bv dr$$

Because every segment of the bar is moving perpendicular to $\mathbf{B}$, an e.m.f. $d\mathcal{E}$ of the same form is generated across each segment. Summing the e.m.f. induced across all segments, which are in series, gives the **total e.m.f.** between the ends of the conducting bar as:

$$\mathcal{E} = \int Bv dr = B \int (r\omega) dr = B\omega \int_0^\ell r dr = B\omega \left[\frac{r^2}{2} \right]_0^\ell = \boxed{\frac{1}{2} B\omega \ell^2}$$

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Example. The structure of **Singapore Flyer** is a gigantic wheel with a diameter of 150 m. There are 28 capsules on the circumference and the position of each capsule is tightened with 4 metal cables to the center. These cables move in the magnetic field of the Earth, so each cable acts like the conducting bar in the previous example. You had a green plan to use the e.m.f. generated by the rotation of the flyer to light a 220-V spotlight on the wheel. Will this idea work if each complete rotation of the wheel requires 30 minutes. (The magnitude of the magnetic field of the Earth is 0.5×10^{-4} T. Assume the best situation that the circular area enclosed by the Singapore flyer is 90° to the Earth magnetic field.)



Answer:

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{30 \times 60 \text{ s}} = 3.4907 \times 10^{-3} / \text{s}$$

$$\text{The total e.m.f. generated by 112 (28x4) cables} = \left(\frac{1}{2} B \omega l^2 \right) \times 112$$

$$= \frac{1}{2} (0.5 \times 10^{-4} \text{ T}) (3.4907 \times 10^{-3} / \text{s}) \left(\frac{150 \text{ m}}{2} \right)^2 \times 112 = \boxed{0.055 \text{ V}}$$

This total induced e.m.f. can't even light a very small light bulb of 1.35 V.

Even if the angular speed of the flyer can be safely accelerated, this plan will not make the environment greener. As the cables must be part of a circuit in order to provide a voltage to the spotlight, the cables must carry a current. Because these current-carrying cables are in a magnetic field of the Earth, a magnetic force is exerted on each cable and the direction of the force is **opposite** to its direction of motion. As a result, the motor of the flyer must supply additional energy to perform work against this magnetic drag force. Finally the operator of **Singapore Flyer** must provide the energy from other source to light the spotlight so nothing is gained for free by this method!



Increase the angular speed of the flyer since we cannot increase the earth magnetic field !!!!!

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Lenz's Law Revisited

- Developed by German physicist Heinrich Lenz.
- Faraday's law $\left(\varepsilon = - \frac{d\Phi_B}{dt} \right)$ indicates that the induced e.m.f. and the change in magnetic flux have opposite algebraic signs.
- This has a physical interpretation by the **Lenz's law**, which states that *"the induced current in a loop is in the direction that creates a magnetic field that opposes the change in magnetic flux through the area enclosed by the loop."*

The Lenz's law means "the induced current tends to keep the original magnetic flux through the circuit from changing".

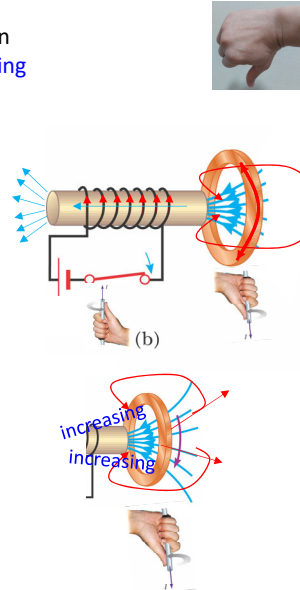
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Example. A metal ring is placed near a solenoid, as shown in figure (a). Find the direction of the induced current in the ring

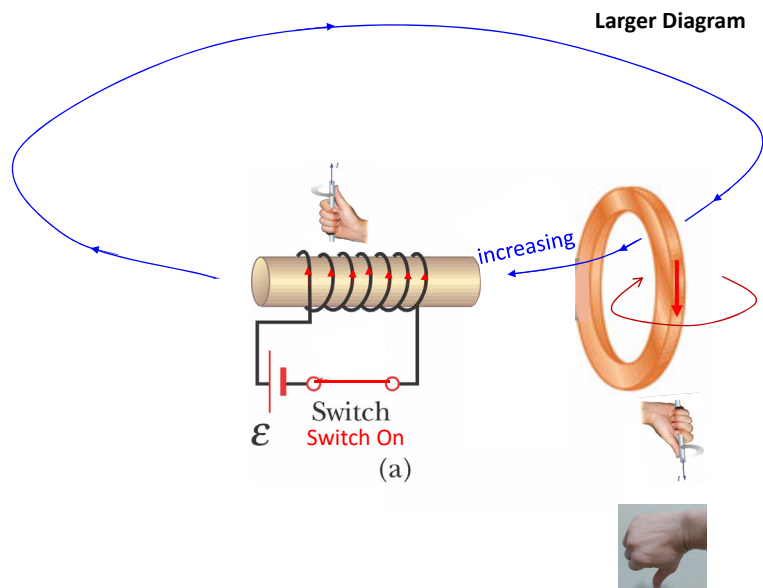
- (i) at the instant the switch in the circuit containing the solenoid is connected

Answer:

At the instant the switch is connected, the situation changes from one in which no magnetic flux exists in the ring to one in which flux exists, and the magnetic field (into the ring) is as shown in figure (b). To counteract this change in the increasing flux, the induced current in the ring must set up a magnetic field directed in the opposite direction, i.e., going out from the ring. This requires a current directed by the right-hand rule.



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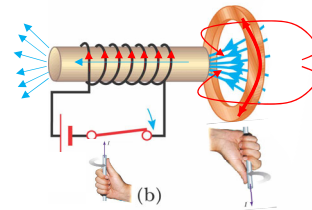
30

Example. A metal ring is placed near a solenoid, as shown in figure (a). Find the direction of the induced current in the ring

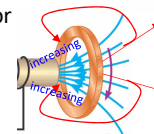
- (i) at the instant the switch in the circuit containing the solenoid is connected

Answer:

At the instant the switch is connected, the situation changes from one in which no magnetic flux exists in the ring to one in which flux exists, and the magnetic field (into the ring) is as shown in figure (b). To counteract this change in the increasing flux, the induced current in the ring must set up a magnetic field directed in the opposite direction, i.e., going out from the ring. This requires a current directed by the right-hand rule.



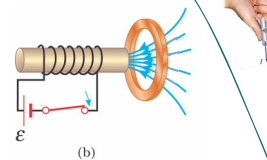
or



- (ii) after the switch has been closed for several seconds

Answer:

After the switch has been closed for several seconds, no change in the magnetic flux through the loop occurs; hence, the induced current in the ring is zero.



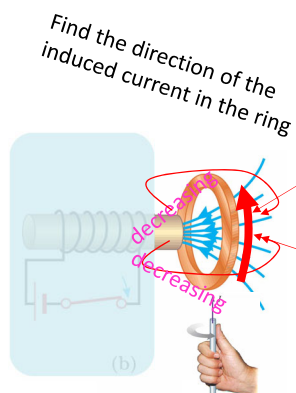
31

- (iii) at the instant the switch is subsequently disconnected.

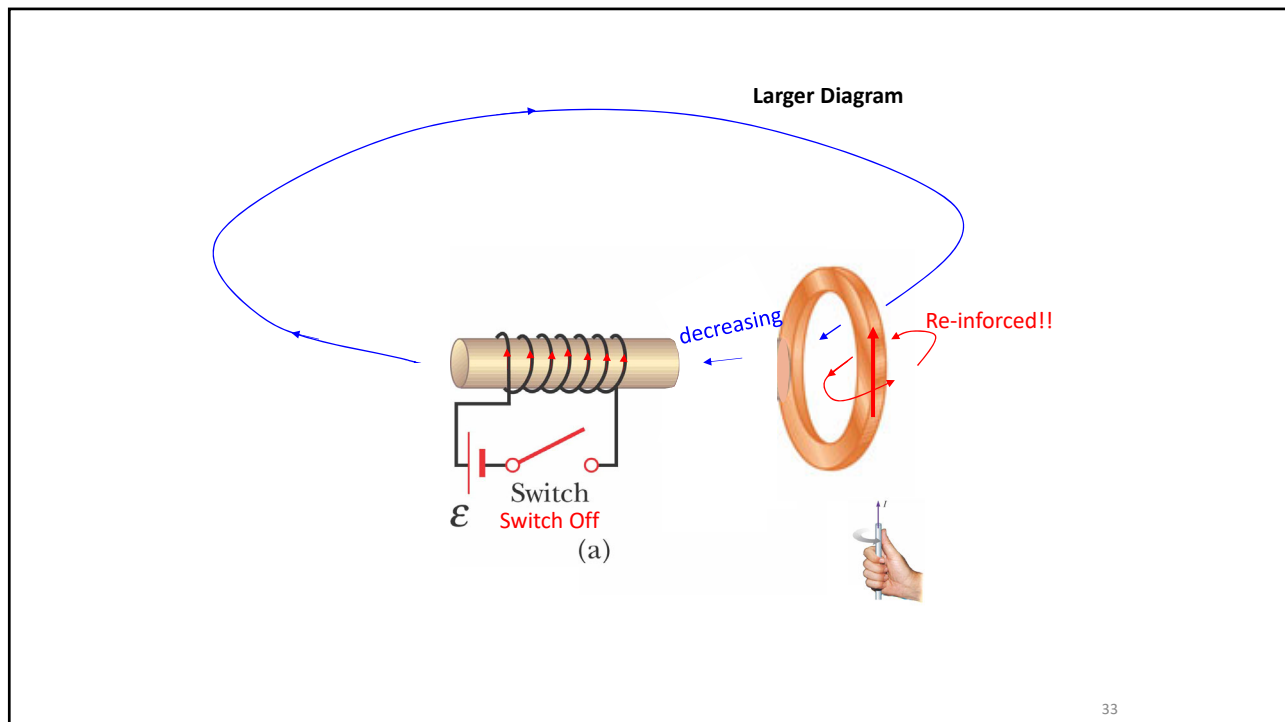
Answer:

Disconnecting the switch changes the situation from one in which magnetic flux exists in the ring to one in which there is no magnetic flux. The direction of the induced current is as shown in figure to counteract the decrease in the flux produced by the solenoid.

(To counter the decrease mean the induced current has to reinforce the previous decreasing magnetic flux to prevent the decrease.)

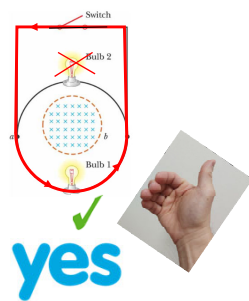


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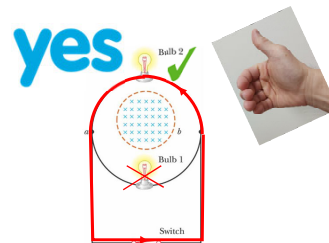


Direction of Induced Current Revisited

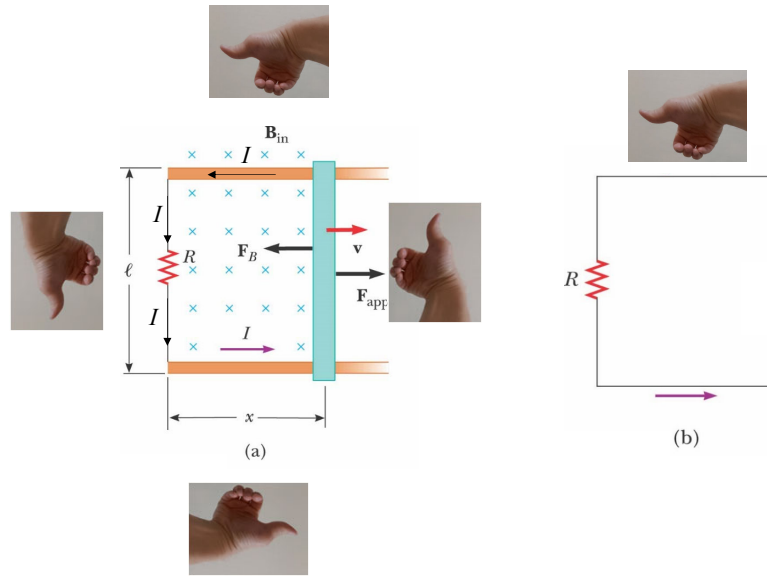
Switch **On**



Switch **On**

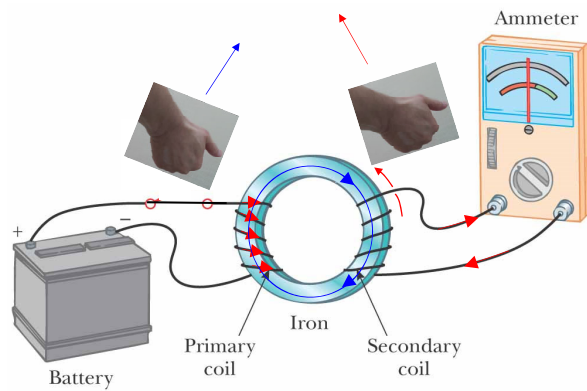


Direction of Induced Current Revisited



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Faraday's Experiment



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Induced e.m.f. and Electric Fields ^(non-conservative)

- This (induced) electric field is created in the conductor as a result of the changing magnetic flux.
- Even in the **absence** of a conducting loop, a changing magnetic field will generate an electric field with the presence of conductor in empty space.
- This **induced** electric field is **non-conservative**

(On other hand, the electric field produced by stationary charges is conservative, so the electric force in that electric field is conservative. The work done by the electric force in a close loop in that electric field produced by stationary charges is 0 because the field is conservative.)

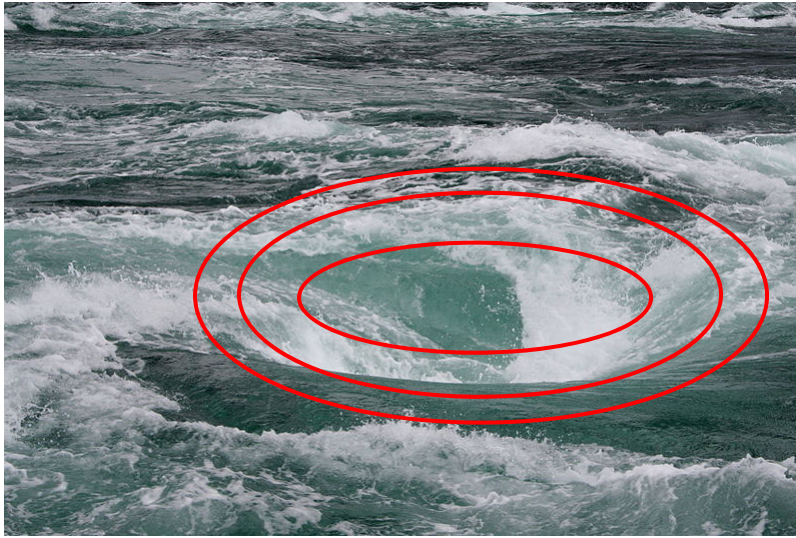
37

Induced emf and Electric Fields, cont.

- From the potential gradient ($\mathbf{E} = -\frac{dV}{ds}$), the **e.m.f. for any closed path can be expressed as the line integral of $\mathbf{E} \cdot d\mathbf{s}$ over the closed path.**
 $\rightarrow \mathbf{E} \cdot d\mathbf{s} = -dV$
- Faraday's law ($|\mathcal{E}| = \left| -\frac{d\Phi_B}{dt} \right|$) can be written in a general form: $\oint (\mathbf{E} \cdot d\mathbf{s}) = \oint (-dV) = \mathcal{E} = -\frac{d\Phi_B}{dt}$
- This induced electric field is a **non-conservative** field generated by a changing magnetic field.
- The induced electric field **cannot** be an electrostatic field because if the field were electrostatic and hence conservative, the line integral of $\mathbf{E} \cdot d\mathbf{s}$ for the closed path would be zero. But $\oint \mathbf{E} \cdot d\mathbf{s} = -\frac{d\Phi_B}{dt} \neq 0$.
 induced emf $\neq 0$

(Two induced (deliberate) injuries do no make a recovery!!)

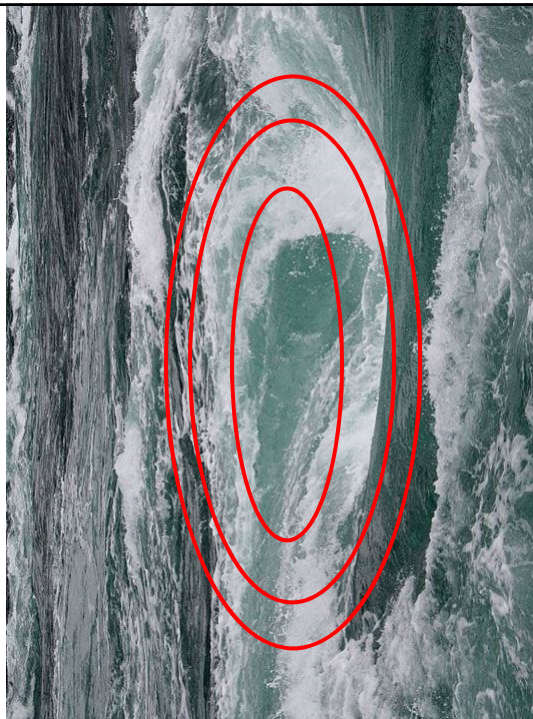
38



Eddy Water Current

Induced electric field

39

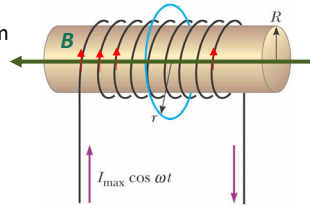


Eddy Water Current

Induced electric field

40

Example. A long solenoid of radius R has n turns of wire per unit length and carries a time-varying current that varies sinusoidally as $I = I_{\max} \cos \omega t$, where $I_{\max}$ is the maximum current and ω is the angular frequency of the alternating current source as shown in the figure.



- (i) Determine the magnitude of the induced electric field outside the solenoid at a distance $r > R$ from its long central axis.

Answer:

$$\oint (E \cdot ds) = -\frac{d\Phi_B}{dt} = \mathcal{E}$$

The magnetic field outside the solenoid is 0 thus the magnetic flux Φ_B enclosed by the circle of radius r , where $r > R$, is $\Phi_B = B \times A = B(\pi R^2)$.

By Faraday's law expressed by line integral,

$$E(2\pi r) = \oint \mathbf{E} \cdot d\mathbf{s} = \mathcal{E} = -\frac{d\Phi_B}{dt} = -\frac{d(B(\pi R^2))}{dt} = -\pi R^2 \left(\frac{dB}{dt} \right) \rightarrow E(2\pi r) = -\pi R^2 \left(\frac{dB}{dt} \right) \dots (1)$$

The magnetic field inside a solenoid is $B = \mu_0 n I = \mu_0 n I_{\max} \cos \omega t$.

$$\begin{aligned} (1) \text{ becomes } &= -\pi R^2 \left(\frac{d(\mu_0 n I_{\max} \cos \omega t)}{dt} \right) = -\pi R^2 \mu_0 n I_{\max} \frac{d(\cos \omega t)}{dt} \\ &= -\pi R^2 \mu_0 n I_{\max} (-\omega \sin \omega t) \end{aligned}$$

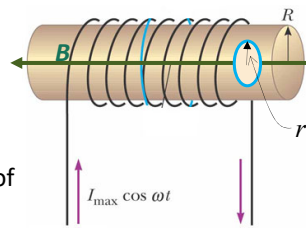
Not conservative !!

$$\rightarrow E = \left(\frac{\mu_0 n I_{\max} \omega R^2}{2} \right) \frac{\sin \omega t}{r} \quad \text{for } r > R.$$

Hence, the amplitude of the induced electric field outside the solenoid falls off as $1/r$ and varies sinusoidally with time.

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- (ii) What is the magnitude of the induced electric field inside the solenoid, a distance r from its axis?



Answer:

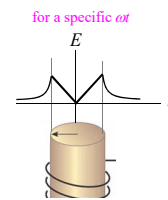
The magnetic flux Φ_B enclosed by the circle of radius r , when $r < R$, is $\Phi_B = B \times A = B(\pi r^2)$.

$$\text{From } E = \left(\frac{\mu_0 n I_{\max} \omega R^2}{2r} \right) \sin \omega t \quad \text{for } r > R$$

$$\begin{aligned} \rightarrow E &= \frac{\mu_0 n I_{\max} \omega r^2}{2r} \sin \omega t \quad \text{for } r < R \\ &= \left(\frac{\mu_0 n I_{\max} \omega}{2} \right) r \sin \omega t \quad \text{for } r < R \end{aligned}$$

(non-conservative)

This shows that an electric field can be induced both inside and outside the solenoid. The amplitude of the induced electric field inside the solenoid due to the changing magnetic flux increases linearly with r and varies sinusoidally with time.



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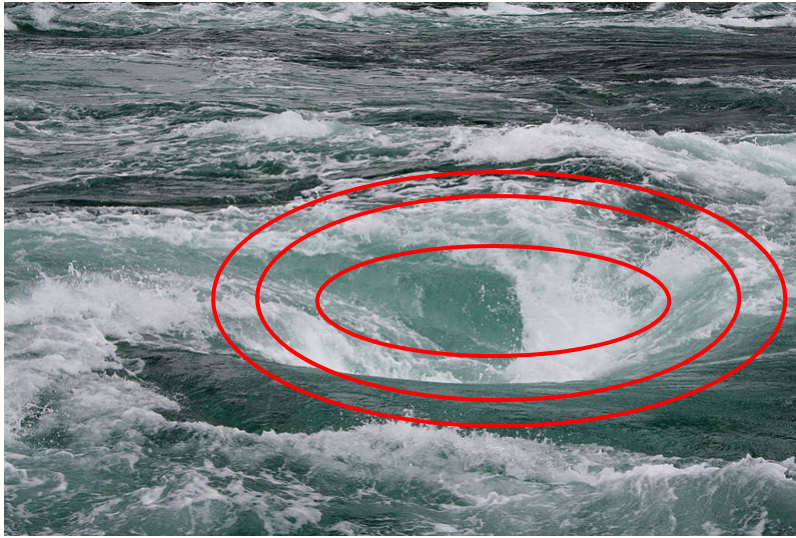
Gas Stove

43

Application:
Induction Stove



44

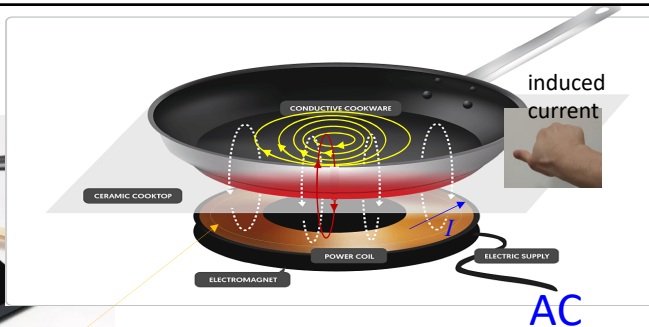


Eddy Water Current

An induction stove cooks food through the use of eddy electric currents.

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Application: Induction Stove



- Metal coil below the cooking surface carries an alternating current which produces an alternating magnetic field. **But Lenz's law opposes.**
- With the regular change of magnetic flux, EMF is induced in metallic pot's bottom surface which leads to an **induced current** also known as **eddy current** in the cooking pot.
- Metal in cooking pot offers a finite resistance which heats up as energy is dissipated.

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Application: Induction Stove



AC

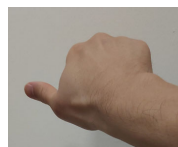


47

Application: Induction Stove

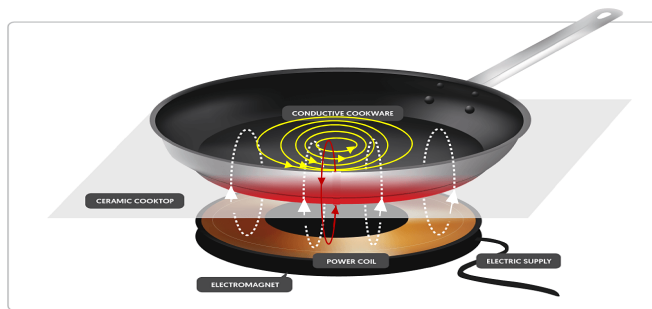


AC



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Application: Induction Stove



AC

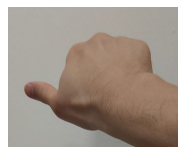


49

Application: Induction Stove



AC



50

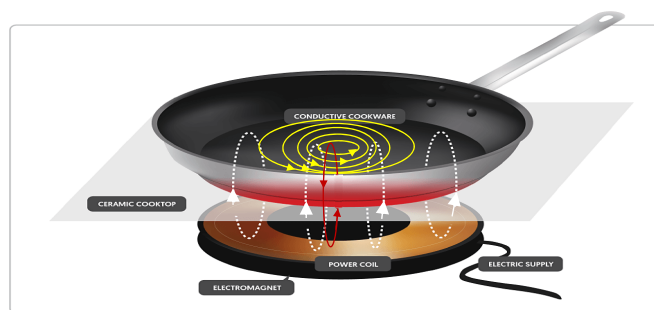


Eddy Water Current

An induction stove cooks food through the use of eddy electric currents.

51

Application: Induction Stove

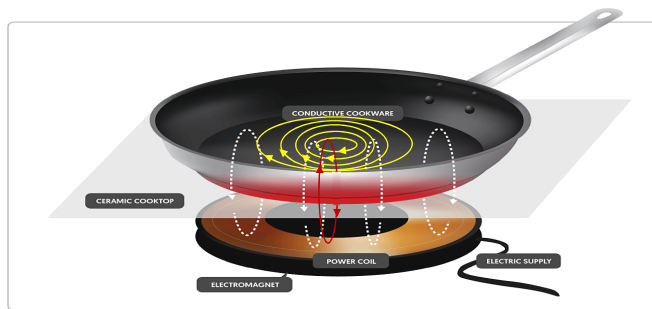


AC



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Application: Induction Stove



AC



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Gas Stove

Never touch the surface of the gas stove because it is burning hot!!!

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Induction Stove

Eddy current (that generates heat) is only in the cooking utensil above the induction stove. Can you touch the surface of the induction stove?

Answer:

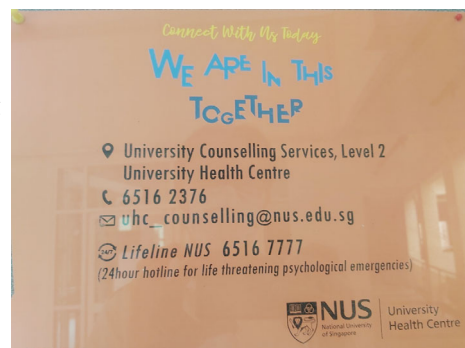
55

Helplines to de-stress

We are not superman/superlady. Sometime and somewhere in our life journey we will need help. Here they are:

Lifeline NUS (for life threatening psychological emergencies):
6516 7777 (24 hours)

6516 2376 (Office hours)
uhc_counselling@nus.edu.sg



Samaritans of Singapore Hotline: 1800 221 4444
Institute of Mental Health's Helpline: 6389 2222
Singapore Association of Mental Health Helpline: 1800 283 7019

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